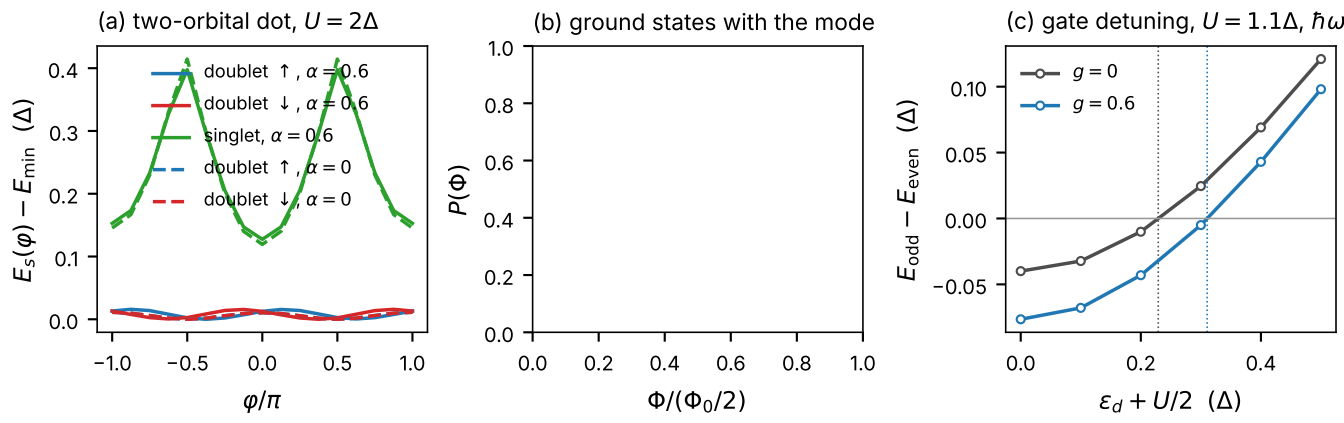
(a) two-orbital dot, $U = 2\Delta$
doublet ↑, $\alpha = 0.6$
doublet ↓, $\alpha = 0.6$
singlet, $\alpha = 0.6$
doublet ↑, $\alpha = 0$
doublet ↓, $\alpha = 0$
$E_s(\varphi) - E_{\min}$ (Δ)
φ/π
(b) ground states with the mode
$P(\Phi)$
$\Phi/(\Phi_0/2)$
(c) gate detuning, $U = 1.1\Delta$, $\hbar\omega$
$g = 0$
$g = 0.6$
$E_{\text{odd}} - E_{\text{even}}$ (Δ)
$\varepsilon_d + U/2$ (Δ)